

The **future-extension** field SHALL be encoded as OCTET STRING and it SHALL be an exact copy of the DER encoded extension field (including the DER encoded ASN.1 OID of the extension) in the corresponding X.509 certificate. These extension fields in a Matter certificate SHALL be encoded in the same order as they appeared in the original X.509 certificate.

6.5.12. Matter certificate Extensions Encoding Rules

The rules that SHALL be followed when encoding the Matter certificate are:

- For a **Matter Node Operational Certificate (NOC)**:
 - The basic constraints extension SHALL be encoded with **is-ca** set to **false**.
 - The key usage extension SHALL be encoded with exactly one flag: **digitalSignature**.
 - The extended key usage extension SHALL be encoded with exactly two **key-purpose-id** values: **serverAuth** and **clientAuth**.
 - The subject key identifier extension SHALL be present.
 - The authority key identifier extension SHALL be present.
- For **Matter ICA Certificate** and **Matter Root CA Certificate**:
 - The basic constraints extension SHALL be encoded with **is-ca** set to **true**.
 - The key usage extension SHALL be encoded with at least the two flags **keyCertSign** and **CRL-Sign** included, and MAY include a third flag: **digitalSignature**. No additional flags are permitted.
 - The extended key usage extension SHALL NOT be present.
 - The subject key identifier extension SHALL be present.
 - The authority key identifier extension SHALL be present.
 - For the **Matter Root CA Certificate** the authority key identifier extension SHALL be equal to the subject key identifier extension.
- For **Vendor ID Verification Signer Certificate**:
 - The basic constraints extension SHALL be encoded with **is-ca** set to **false**.
 - The key usage extension SHALL be encoded with exactly one flag: **digitalSignature**.
 - The subject key identifier extension SHALL be present.
 - The authority key identifier extension SHALL be present.
- The extensions SHALL appear in the same order in the Matter certificate and in the corresponding X.509 certificates.
- For a Matter Firmware Signing Certificate these rules SHALL be followed:
 - The basic constraints extension SHALL be encoded with **is-ca** set to **false**.
 - The key usage extension SHALL be encoded with exactly one flag: **digitalSignature**.
 - The extended key usage extension SHALL be encoded with exactly one **key-purpose-id** values: **codeSigning**.
 - The subject key identifier extension SHALL be present.